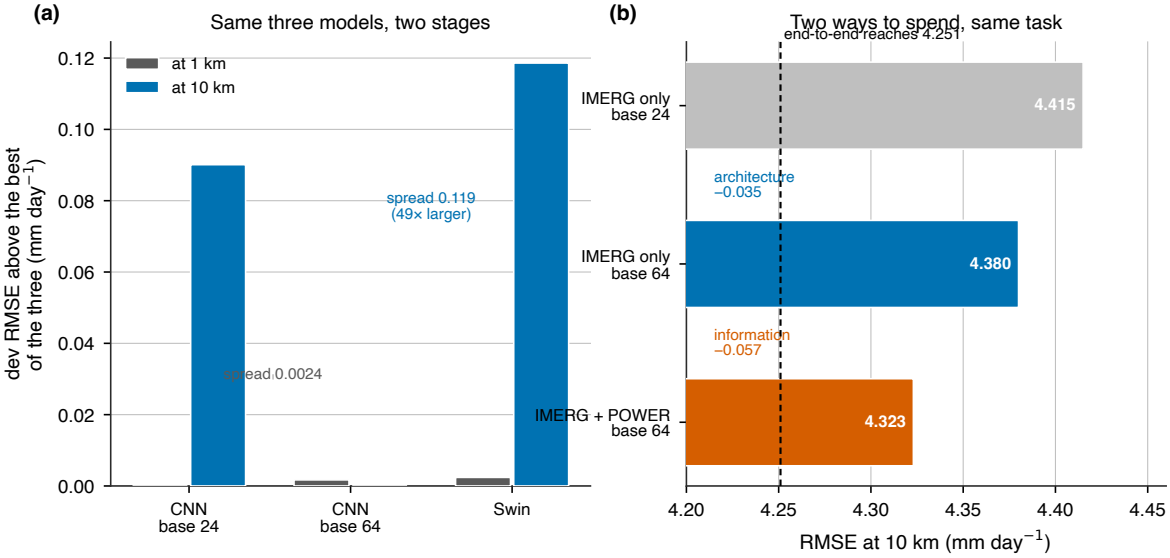
(a)
Same three models, two stages
at 1 km
at 10 km
dev RMSE above the best of the three (mm day⁻¹)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
spread 0.119
(49× larger)
spread 0.0024
CNN base 24
CNN base 64
Swin
(b)
Two ways to spend, same task
end-to-end reaches 4.251
IMERG only base 24
4.415
architecture −0.035
IMERG only base 64
4.380
information −0.057
IMERG + POWER base 64
4.323
4.20
4.25
4.30
4.35
4.40
4.45
RMSE at 10 km (mm day⁻¹)